

Dynamics of Mitochondrial Transfer in Tumors

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Abstract

Mitochondrial transfer within the tumor microenvironment (TME) represents a key indicator of immune evasion and cancer progression(1)(2)(3). Recent studies have demonstrated that cancer cells can acquire functional mitochondria from immune cells to restore oxidative phosphorylation (OXPHOS) and promote survival(4)(5). In contrast, other reports describe transfer in the opposite direction, where tumor-derived mutated mitochondria are delivered to immune cells, making immune cells dysfunctional(3). This highlights a complex mode of mitochondrial transfer within the TME.

In our analysis, we aim to infer mitochondrial transfer by examining mixed mitochondrial variant signatures. We replicated the findings of Zhang *et al.*(5) using their computational tool MERCI(5) to investigate mitochondrial exchange between tumor cells and T cells. Furthermore, we have identified suitable datasets to further explore mitochondrial transfer using MERCI(6). These findings will contribute to a growing understanding of how intercellular crosstalk shapes tumor-immune interactions and may reveal new vulnerabilities for therapeutic targeting and prognosis(2).

Introduction

Mitochondria are best known for orchestrating oxidative phosphorylation, yet their functional repertoire extends well beyond ATP production to include regulation of apoptosis, reactive oxygen species balance, and innate immune signaling(1). Over the past decade it has become evident that mitochondria are not confined to the cytoplasm of their cell of origin. Instead, organelles or mitochondrial DNA fragments can be horizontally transferred between neighboring cells through tunneling nanotubes, gap junction-like structures, cell fusion events, and extracellular vesicles loaded with mitochondrial cargo(2),(4).

Solid tumors are densely populated ecosystems in which malignant, stromal, and immune cell populations compete for oxygen and nutrients. Within this milieu, mitochondrial transfer has been implicated as a metabolic barter system: stressed cancer cells lacking intact oxidative phosphorylation can acquire healthy mitochondria from infiltrating immune cells(4),(5), while damaged or mutated mitochondrial genomes may traffic in the opposite direction, thereby attenuating immune surveillance(3). Understanding the prevalence, directionality, and functional impact of such transfers across tumor types and stages is essential because they may explain heterogeneity in therapeutic response to immune check-

point blockade, chimeric antigen receptor T-cell therapy, and metabolic inhibitors(2),(3).

First, we seek to replicate and extend the MERCI benchmark analysis to validate our computational pipeline(5). Second, we quantify mitochondrial coverage and variant burden to establish detection sensitivity across single-cell RNA sequencing platforms. Third, we interrogate donor-receiver dynamics in coculture systems representative of early versus advanced disease(7). Finally, we outline a roadmap for scaling these analyses to publicly available datasets spanning multiple cancer types and integrating single-cell multi-omics.

Results

Replication of MERCI Benchmarking Analysis

We successfully replicated the benchmark analysis from the MERCI paper using their publicly available code and dataset (5). The MERCI (Mitochondrial-Enabled Reconstruction of Cellular Interactions) method is a computational tool that traces mitochondrial transfer between T cells and cancer cells using single-cell RNA sequencing data.

The analysis focused on the Bench1 dataset, which contains cocultured (CC) cancer cells with transferred mitochondria, monocultured (MC) cancer cells without foreign mitochondria, and T cells as the donor population. MERCI uses two complementary approaches to identify mitochondrial receiver cells: DNA-based detection using mitochondrial single nucleotide variants (mtSNVs) and RNA-based estimation of the relative abundance of transferred mitochondria.

First, we analyzed the coverage of the mitochondrial genome in the sample. Figure 1 illustrates the read coverage distribution across different sequencing depths (1x to 100x) for the mitochondrial genome. The proportion of mitochondrial genome covered decreases with increasing read depth requirements. At lower coverage thresholds (1x-10x), a substantial proportion of the mitochondrial genome is captured, but as we require higher coverage (50x-100x), fewer regions meet this stringency. This coverage analysis is critical for quality control in single-cell mitochondrial studies, as sufficient coverage is necessary to reliably detect mtSNVs that serve as genetic barcodes for tracing mitochondrial transfer. The decreasing trend indicates the inherent challenge of achieving deep, uniform mitochondrial genome coverage in standard 10x Genomics scRNA-seq data, which motivated the development of specialized protocols like mtscATAC-seq for improved mitochondrial DNA profiling.

Figure 2 shows the distribution of mitochondrial weight

metrics in the different cells. The MT-Donor-index represents the relative abundance or index score of donor (T cell)-derived mitochondrial DNA variants in each cell. The MT-Receiver-index indicates the likelihood score of a cell being a mitochondrial receiver based on the presence of T cell-enriched mtSNVs. The boxplots reveal that cocultured cancer cells show intermediate values between pure T cells and monocultured cancer cells, providing evidence of mitochondrial transfer. MT-Donor-frac and MT-Receiver-frac, which represent the estimated fractional composition of mitochondria from different cellular origins. MT-Receiver-frac shows higher values in cocultured cancer cells compared to monocultured cells, confirming that cancer cells acquire T cell mitochondria during coculture. This unidirectional transfer from T cells to cancer cells represents a mechanism by which tumors may evade immune surveillance while simultaneously depleting immune cell energy resources.

Figure 3

MERCI shows heterogenous patterns of transfer in both co-cultured samples

We used MERCI to detect mitochondrial transfer in public scRNA-seq datasets. We analyzed two co-culture samples (SRR15339529 and SRR15339531), consisting of macrophages and MDA-MB-231 breast cancer cells.

Dimensionality reduction using UMAP demonstrated that the two dominant cell populations—macrophages and MDA-MB-231 tumor cells—formed well-separated clusters in both samples. This confirms that the preprocessing steps (SRA conversion, FASTQ validation, and Cell Ranger alignment) accurately preserved cell identity and that downstream mt-SNV inference can be interpreted within biologically meaningful cell clusters.

In SRR15339529, the macrophage cluster and the tumor cell cluster were tightly compact and distinct, with no evidence of major batch-driven distortions. SRR15339531 showed a similar topology, though the cancer-cell cluster was larger and more heterogeneous, indicating sample-specific differences in cell capture or transcriptional variation.

MERCI’s receiver-flag annotation—indicating cells enriched for donor-derived mitochondrial variants—revealed markedly different transfer phenotypes between the two samples. In SRR15339529, receiver-positive cells were almost exclusively localized within the macrophage cluster, with MDA-MB-231 cells showing minimal receiver signal. This pattern strongly suggests directional mitochondrial transfer from MDA-MB-231 cells to macrophages. Conversely, receiver-positive cells were distributed across both macrophages and tumor cells, producing a more heterogeneous transfer landscape. This might suggest bidirectional or less constrained mitochondrial exchange.

These results demonstrate that MERCI is sensitive to sample-specific differences in mitochondrial communication that are not apparent from transcriptomic clustering alone.

Rcm distributions indicate enriched mitochondrial content in receiver cells

To further assess mitochondrial transfer, MERCI also evaluates Rcm (Relative copy number of mitochondria). In SRR15339529, elevated Rcm values were concentrated within a subset of cells corresponding to receiver-positive macrophages, supporting a targeted transfer event. The shape of the Rcm distribution—showing peaks in the mid-to-upper deciles—indicates that these cells experienced a pronounced increase in mitochondrial content relative to the population baseline.

In SRR15339531, Rcm elevations were more evenly distributed across deciles, matching the more dispersed receiver-flag pattern. This suggests a broad mitochondrial exchange process. The alignment of Rcm patterns with UMAP-based receiver assignments supports the robustness of MERCI’s mtSNV-based detection approach.

MT weight metrics strengthen evidence of transfer

MERCI computes two additional metrics (MT Donor index and MT Donor frac) which quantify the contribution of donor-specific mitochondrial variants within each cell. Across both samples, receiver-flagged cells showed elevated donor-variant contributions, reflected by upward shifts in both MT weight distributions. Although the magnitude varied between samples, the consistent increase in donor-specific signal across receiver-positive cells provides validation of MERCI’s inference.

Methods

This study implements a complete computational workflow to convert raw Sequence Read Archive (SRA) files into high-quality aligned BAM files using FASTQ extraction and the Cell Ranger processing pipeline. The overall workflow consists of three major stages:

1. SRA acquisition and verification
2. FASTQ generation and quality validation
3. Alignment, barcode processing, and BAM generation using Cell Ranger

Each step, along with its theoretical basis, input-output structure, and algorithmic logic is described below.

A. SRA Acquisition and Storage

Input: SRA accession identifiers (SRR15339529, SRR15339530, SRR15339531, SRR15339532). **Output:** Local SRA files stored in `sra_files/`.

Two complementary methods were used for retrieval:

- Direct object download from the NCBI AWS S3 bucket
- Fallback download using the NCBI `prefetch` utility

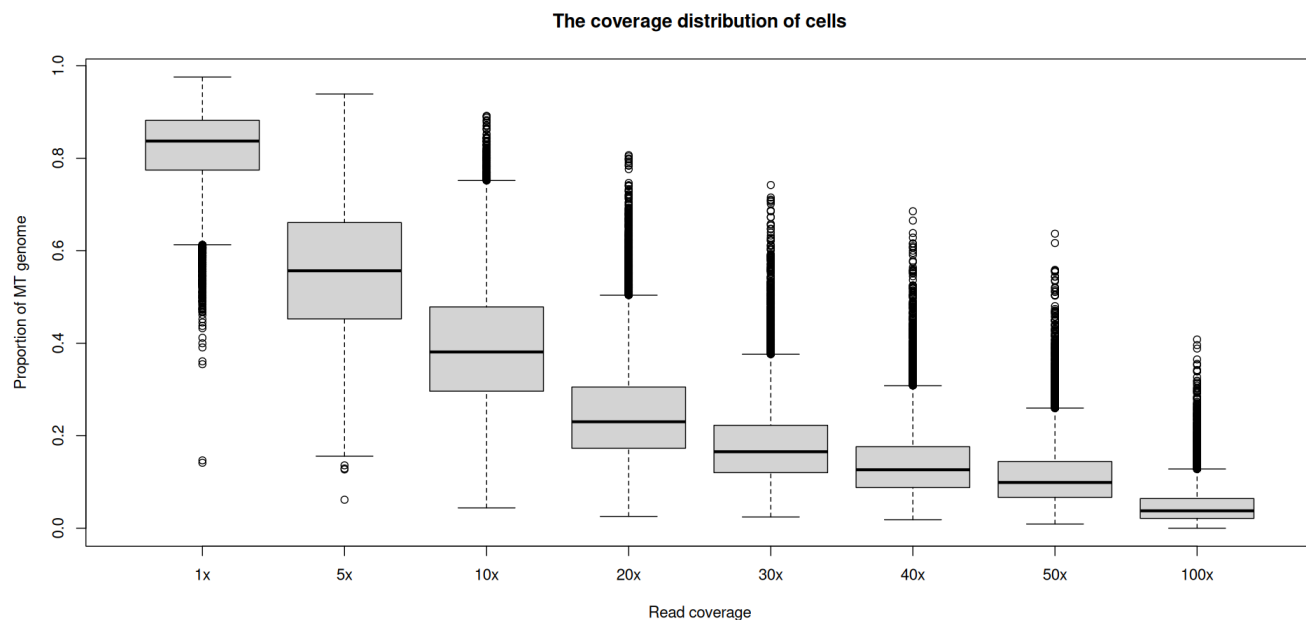


Figure 1. Boxplots showing the distribution of read coverage per cell scRNA-seq dataset (Bench1). Box center line: median; box limits: upper and lower quantiles;

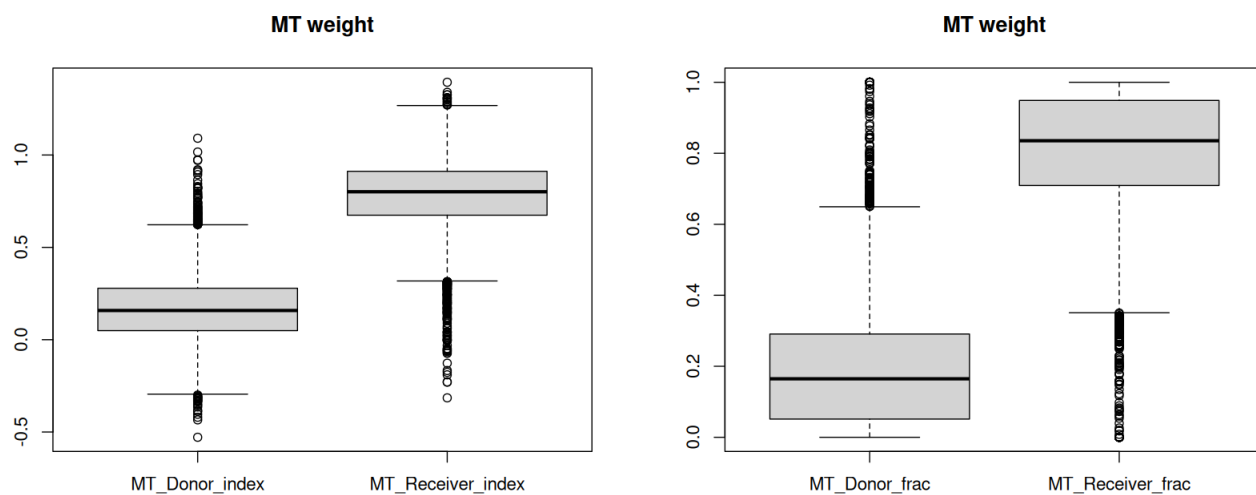
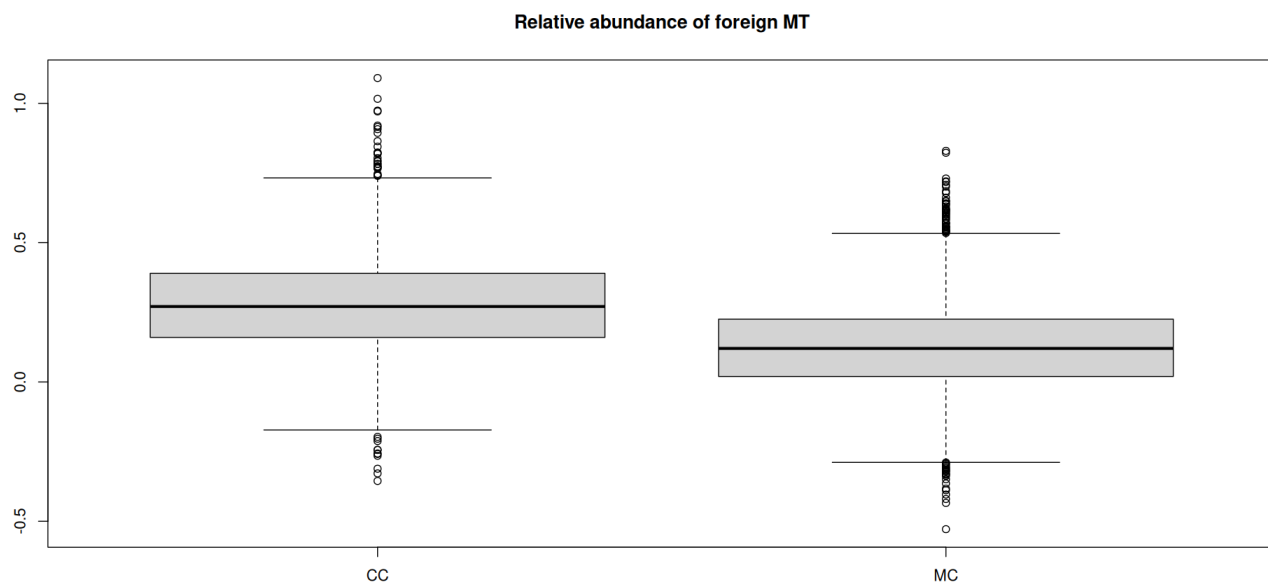


Figure 2. Plots distribution of mitochondrial weight metrics across cells (Bench1). Box center line: median; box limits: upper and lower quantiles; box whiskers: 1.5×



UMAP colored by cell_type

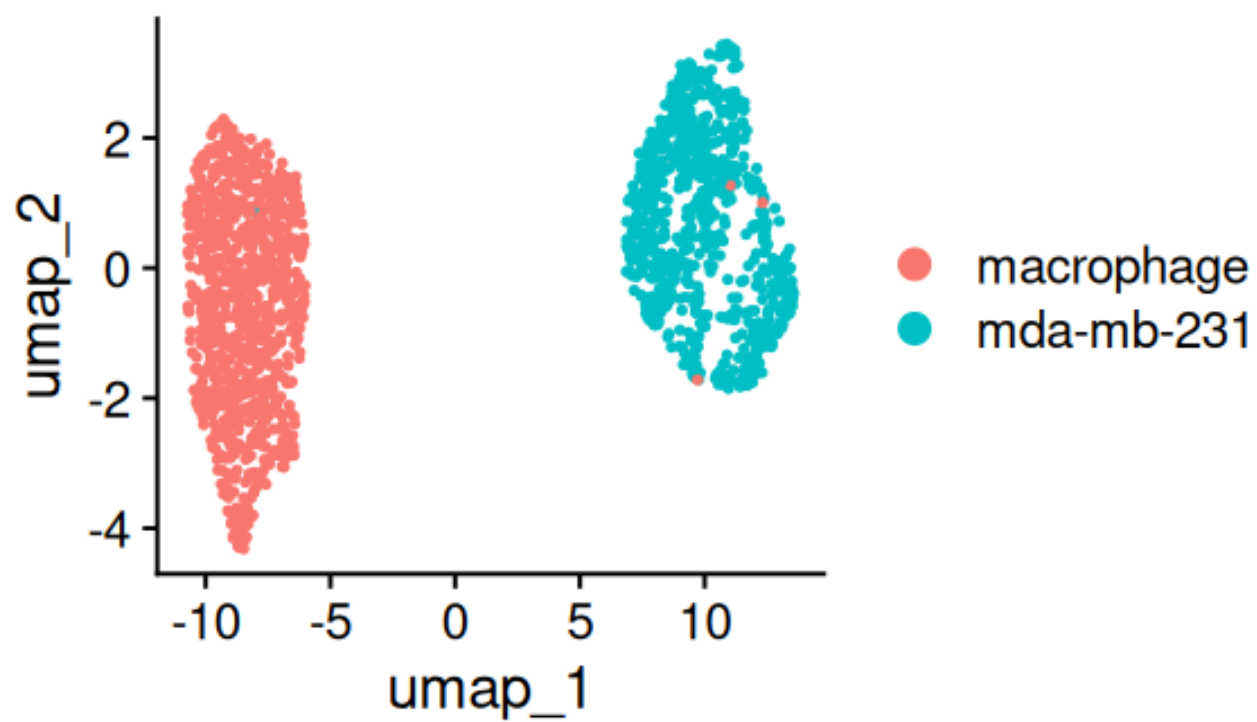


Figure 4. UMAP showing clusters, coloured by cell type, SRR15339529

UMAP colored by cell_type

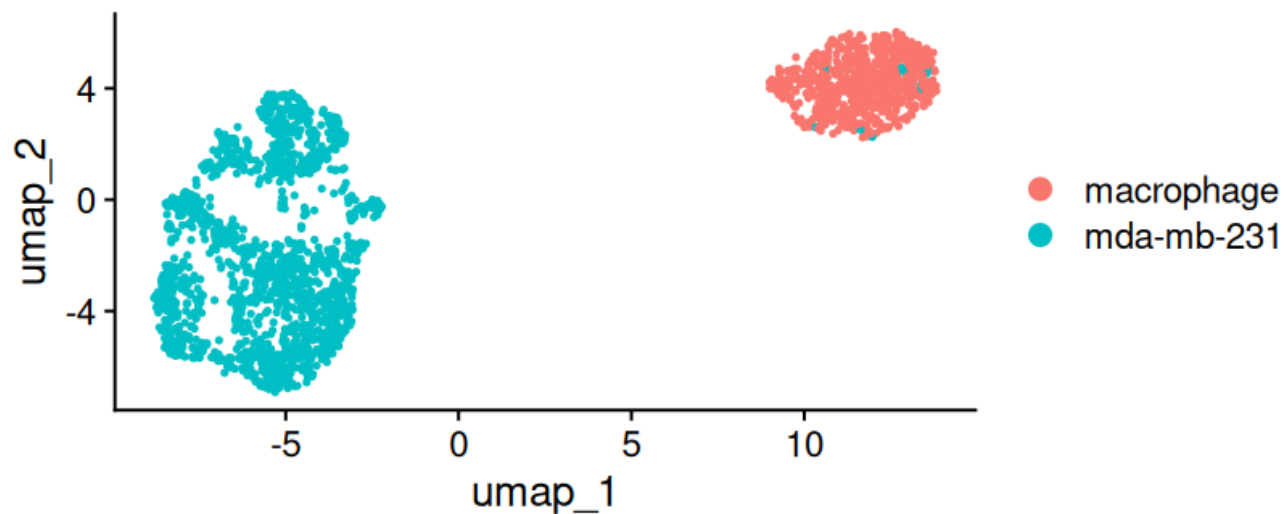


Figure 5. UMAP showing clusters, coloured by cell type, SRR15339531

UMAP colored by receiver_flag

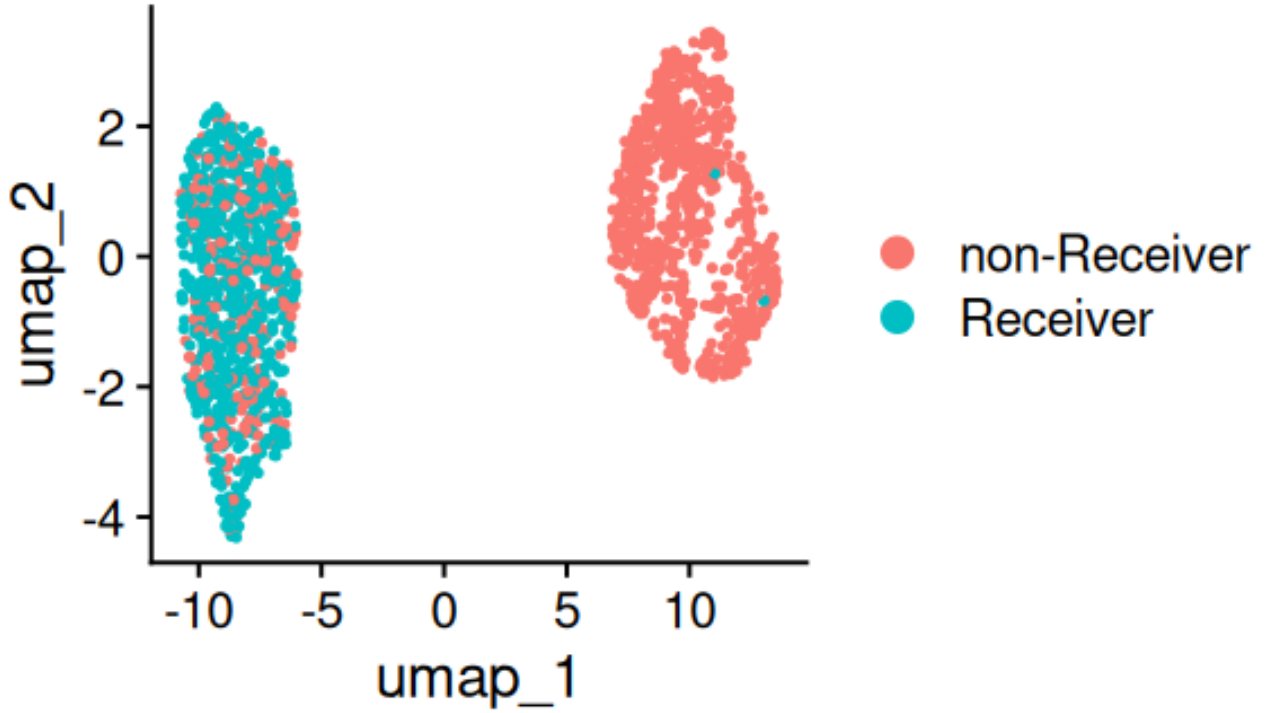


Figure 6. UMAP showing clusters, coloured by receiver flag, SRR15339529

The SRA format stores sequencing reads in a compressed, indexed container. The retrieval step ensures a complete bit-wise copy of the sequencing run before downstream extraction.

B. Verification of Required Tools

Before conversion, the following command-line tools were validated:

- `prefetch` — SRA retrieval and verification
- `fastq-dump` — conversion of SRA to FASTQ
- Conda environment activation (`conda activate`)

The pipeline exits if any required tool is unavailable, ensuring reproducibility.

C. Conversion of SRA to FASTQ

Input: Raw SRA file **Output:** FASTQ files *paired reads*: `sample_1.fastq.gz`, `sample_2.fastq.gz`
unpaired reads: `sample.fastq.gz` (optional)

The FASTQ format encodes each sequencing read as four lines:

$$\text{FASTQ read block} = \begin{cases} @SEQID & (\text{identifier}) \\ ACTG \dots & (\text{nucleotide sequence}) \\ + & (\text{separator}) \\ IIII \dots & (\text{Phred quality encoding}) \end{cases}$$

Each read contributes four lines; therefore the approximate number of reads is

$$N_{\text{reads}} = \frac{L}{4},$$

where L is the total number of lines in the FASTQ file.

The `fastq-dump` algorithm performs:

1. decompression of the SRA container;
2. demultiplexing into paired-end streams;
3. optional removal of technical reads;
4. gzip compression of FASTQ outputs.

The `-split-3` flag ensures proper treatment of paired and orphan reads.

UMAP colored by receiver_flag

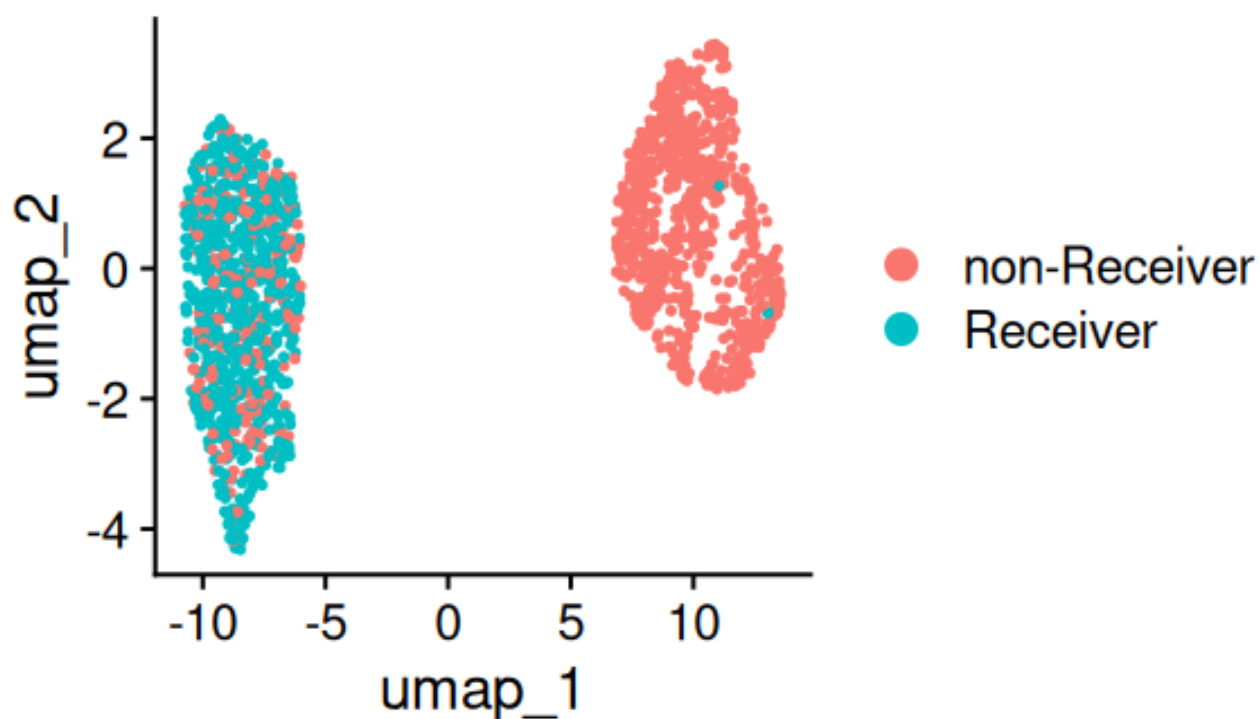


Figure 7. UMAP showing clusters, coloured by receiver flag, SRR15339531

SRR15339529

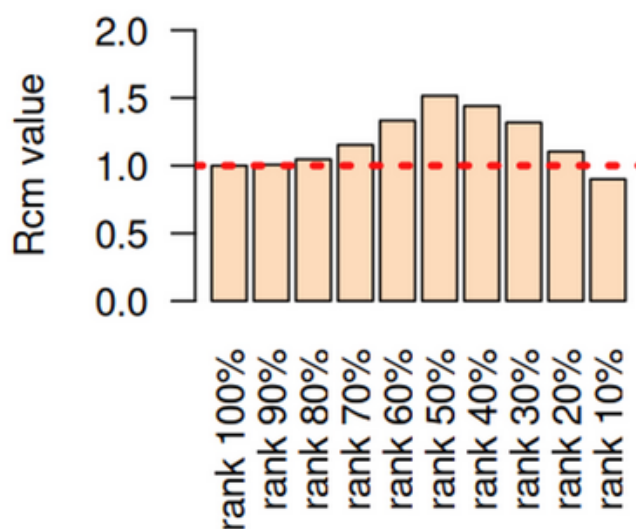


Figure 8. rcm, SRR15339529

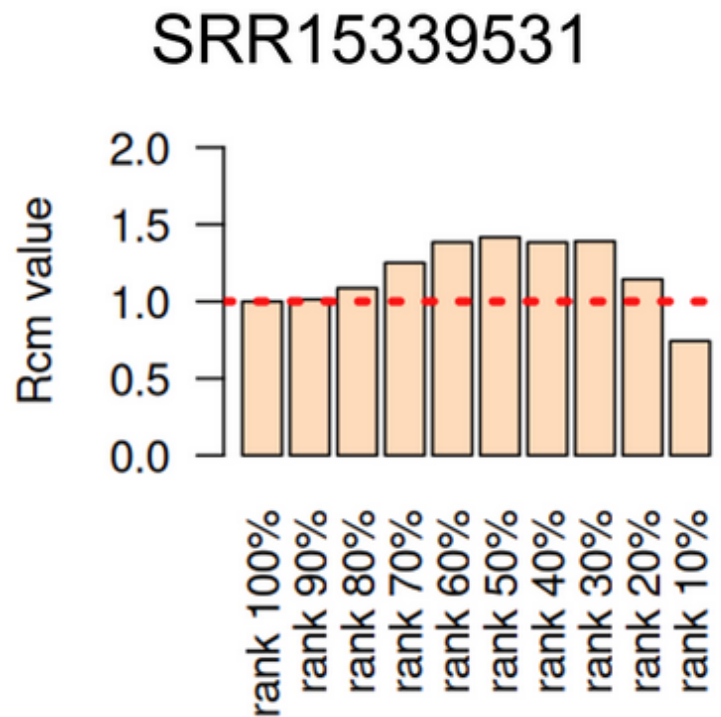


Figure 9. Rcm, SRR15339531

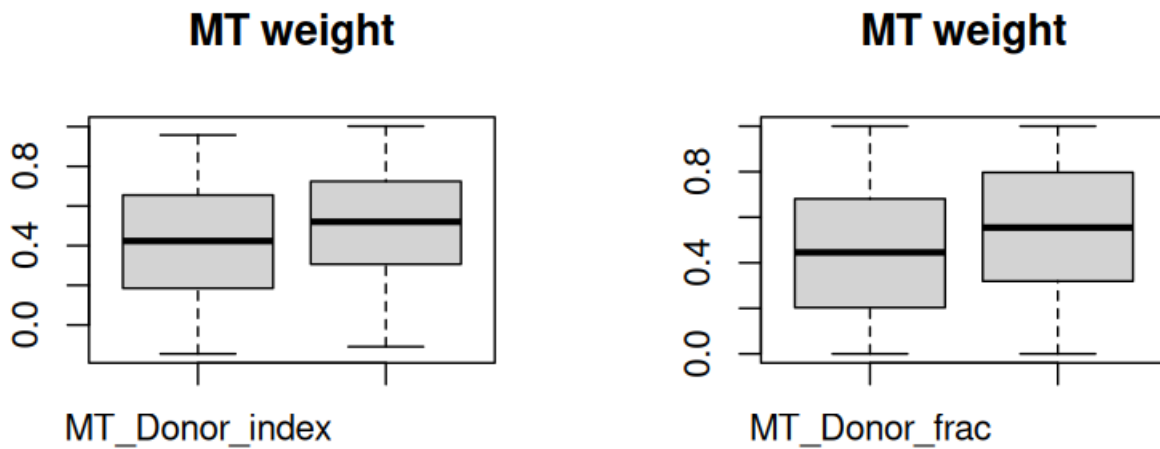


Figure 10. MT donor index, MT donor fraction, SRR15339529

D. FASTQ Quality Validation

Input: FASTQ files **Output:** confirmation of file size, read count, and structure.

Quality checks include:

- verifying gzip file size using `ls -lh`
- computing approximate read count using

$$N_{\text{reads}} = \frac{1}{4} \text{wc -l}(\text{fastq})$$

- manual inspection of the first read using `head -4`

These checks ensure successful SRA extraction and detect early corruption.

E. Renaming FASTQ Files for Cell Ranger Compatibility

The Cell Ranger pipeline requires the following file naming structure:

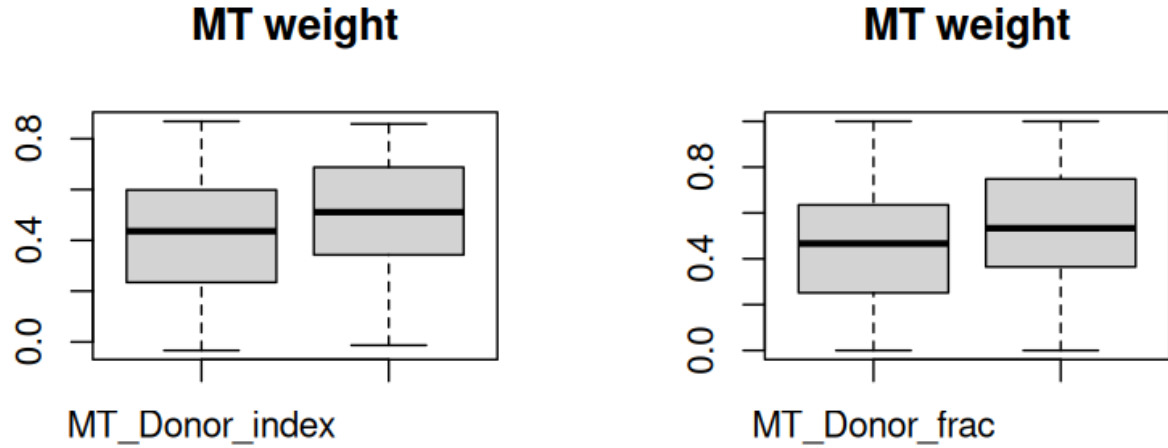


Figure 11. MT donor index, MT donor fraction, SRR15339531

```
<sample>_S1_L001_R1_001.fastq.gz
<sample>_S1_L001_R2_001.fastq.gz
<sample>_S1_L001_I1_001.fastq.gz
```

where:

- R1 = cDNA read
- R2 = UMI/barcode read
- I1 = index read

Correct naming ensures automatic recognition by Cell Ranger.

F. Reference Genome Preparation

Input: 10x Genomics reference “GRCh38-2024-A”

Output: Indexed reference for alignment.

The reference contains:

- genome FASTA
- GTF annotation file
- STAR index generated internally by Cell Ranger

The reference defines valid transcript and gene coordinates used for alignment.

G. Alignment and Quantification Using Cell Ranger

Input: Renamed FASTQ files + 10x reference **Output:**

Sorted, indexed BAM file:

```
possorted_genome_bam.bam
```

Cell Ranger internal steps:

1. **Read alignment:** Cell Ranger uses the STAR aligner, mapping each read to the reference genome. Mathematically, alignment selects the mapping:

$$\hat{x} = \arg \max_{x \in G} S(\text{read}, x)$$

where S is a scoring function based on match/mismatch penalties.

2. **Barcode correction:** Cell barcodes are corrected to nearest valid barcode using Hamming distance:

$$d_H(b_{\text{read}}, b_{\text{valid}}) \leq 1$$

3. **UMI collapsing:** UMI duplicates are removed by grouping identical barcodes/UMIs:

$$\text{UMI}_{\text{unique}} = \text{unique}(b, u, g)$$

where g is gene assignment.

The final BAM file contains:

- aligned reads
- corrected barcodes
- UMI information
- quality metrics in the BAM tags

H. Final Output

The pipeline outputs one BAM file per sample:

```
$PROJECT_DIR/<sample>_bamrun/outs /possorted_genome_bam.bam
```

This BAM serves as the input for downstream single-cell expression analysis.

MERCI Computational Framework

We utilized the MERCI (Mitochondrial-Enabled Reconstruction of Cellular Interactions) computational framework to analyze mitochondrial transfer dynamics between T cells and cancer cells. MERCI is a specialized tool designed to trace mitochondrial transfer at single-cell resolution using sequencing data. The framework consists of two main modules: MERCI-mtSNP for calling mitochondrial single nucleotide variants (mtSNVs) from sequencing data, and the MERCI R package for predicting mitochondrial receiver cells and estimating their mitochondrial composition.

Their data set contains single cell RNA sequencing data from three experimental groups: cocultured cancer cells and CD8+ T cells serving as donors (CC group), monocultured cancer cells without exposure to T cells (MC group). The coculture system allows T cells to transfer mitochondria to cancer cells through mechanisms such as tunneling nanotubes, providing an ideal experimental setup to study mitochondrial transfer dynamics.

MERCI-mtSNP is written in Python 3 and requires the following dependencies: pandas, numpy, pysam, and matplotlib. We ensured all dependency modules were installed before running the pipeline. The MERCI R package was installed following the instructions provided in the GitHub repository.

Dataset Acquisition and Description

We obtained the benchmark dataset (Bench1) from the MERCI GitHub repository. This dataset contains single-cell RNA sequencing data from three experimental groups: cocultured cancer cells and CD8+ T cells (CC group), monocultured cancer cells without T cell exposure (MC group), and primary CD8+ T cells serving as the donor population. The coculture system allows T cells to transfer mitochondria to cancer cells through mechanisms such as tunneling nanotubes, providing an ideal experimental setup to study mitochondrial transfer dynamics. The dataset was generated using 10x Genomics single-cell RNA sequencing technology, which captures both transcriptomic information and mitochondrial DNA variants at single-cell resolution.

Raw sequencing runs for benchmark and exploratory datasets were retrieved from the NCBI Sequence Read Archive. Each .sra file was converted to paired-end FASTQ using the SRA Toolkit's fasterq-dump utility, ensuring lossless reconstruction of read identifiers essential for cell barcode parsing. Reads were aligned against the GRCh38 reference genome with HISAT2 under single-cell mode to preserve cellular barcodes and unique molecular identifiers in the resulting SAM output. Subsequent BAM conversion, sorting, and indexing were executed with samtools to furnish coordinate-sorted alignments compatible with MERCI's mtSNP extractor. During alignment, parameters were tuned to penalize soft clipping and ambiguous mapping, thereby reducing spurious assignments of nuclear mitochondrial DNA segments (NUMTs) that could inflate variant calls.

MERCI-mtSNP: Mitochondrial Variant Calling

We processed the raw sequencing data using the MERCI-mtSNP pipeline to identify mitochondrial single nucleotide variants. The pipeline accepts BAM files (output from Cell Ranger or similar alignment software), the genome reference FASTA file, and cell barcode information. We used the following key parameters specific to 10x scRNA-seq data.

- **-bam-list** : path to a plain text file containing one BAMs produced by Cell Ranger or equivalent. MERCI uses these aligned reads to generate per-cell mitochondrial pileups and call mtSNVs.
- **-reference** : reference genome in FASTA.
- **-barcodes** : 10x cell barcode file (TSV/CSV) listing valid cell barcodes or mapping barcodes to sample/cell IDs
- **-outdir** : output directory
- **-min-coverage** : minimum read depth at a site required to consider variant evidence (adjust based on your coverage QC).
- **-min-af** : minimum allele fraction (allele frequency) required to report a variant in a cell
- **-platform** : optional flag to enable platform-specific handling (set to 10x to account for UMI/barcode formats and read structure typical of 10x Genomics data).

The MERCI-mtSNP module identifies mtSNVs that are enriched in specific cell populations (e.g., T cells versus cancer cells), which serve as genetic barcodes for tracing mitochondrial origins.

MERCI R Package: Receiver Cell Prediction

Following variant calling, we used the MERCI R package to predict which cells are mitochondrial receivers and estimate their mitochondrial composition. The pipeline employs two complementary approaches

DNA-based detection : This approach calculates DNA rank scores based on the presence of T cell-enriched mtSNVs in cancer cells. Cells harboring donor-specific variants are identified as potential mitochondrial receivers.

RNA-based estimation : The pipeline uses support vector regression to estimate the relative abundance of transferred mitochondria based on mitochondrial gene expression patterns. This approach quantifies the fractional composition of mitochondria from different cellular sources.

MT_Donor_index The relative abundance or index score of donor (T cell)-derived mitochondrial DNA variants.

MT_Receiver_index The likelihood score that a cell is a mitochondrial receiver based on T cell-enriched mtSNVs.

MT_Donor_frac The estimated fractional composition of mitochondria in a cell that are donor-derived.

MT_Receiver_frac The estimated fractional composition indicating the proportion of foreign (donor) mitochondria in receiver cells.

CellRanger: Generating BAMs and Barcodes

Discussion

In this study, we applied MERCI to publicly available scRNA-seq co-culture datasets. Across both samples, MERCI successfully identified receiver-positive cells and quantified donor-mitochondrial contributions using a combination of receiver-flags, MT weight metrics, and Rcm distributions.

MERCI successfully detected mitochondrial transfer events in both datasets, but the inferred transfer patterns differed substantially between samples. In SRR15339529, receiver-flagged cells were almost exclusively confined to the macrophage cluster, indicating a highly specific and directional transfer from MDA-MB-231 cells to macrophages. In contrast, SRR15339531 displayed a more diffuse receiver distribution spanning both macrophages and cancer cells. The broader spread of receiver-positive cells, together with more uniformly elevated Rcm values, suggests that mitochondrial exchange in this sample may be influenced by shifts in microenvironmental conditions, increased cellular stress, or differences in cell viability or co-culture dynamics. These observations emphasize MERCI's sensitivity to biologically meaningful heterogeneity that would not be visible from gene expression clustering alone.

The present findings suggest several important avenues for future work. First, combining MERCI with single-cell ATAC-seq unlocks the capacity to correlate mitochondrial genotype with chromatin accessibility, thereby probing whether donor mitochondria reconfigure nuclear transcriptional programs in recipient cells. Second, spatial transcriptomics, particularly technologies capable of resolving sub-cellular localization, will situate transfer events within the architectural blueprint of the TME, determining whether proximity to vasculature or hypoxic gradients modulates transfer propensity. Third, datasets with higher mitochondrial coverage—such as those generated using mtRNA enrichment protocols or specialized platforms like MAESTER—would allow more precise quantification of donor-specific variants and improved estimation of MT weights. Fourth, integrating MERCI-derived mitochondrial transfer scores with clinical or phenotypic metadata could help uncover associations between mitochondrial exchange and tumor grade, disease progression, or patient prognosis. Finally, expanding analyses to additional cell types and co-culture systems would help determine whether the sample-specific differences observed here represent generalizable biological patterns or context-dependent phenomena. Overall, our results demonstrate that MERCI provides a powerful framework for detecting and quantifying mitochondrial transfer using publicly available scRNA-seq data. Although technical and biological limitations remain, the ability to infer transfer at single-cell resolu-

tion opens the door to broader investigations into how mitochondrial exchange shapes tumor-immune interactions and disease progression.

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MERCI Rough Workflow Overview

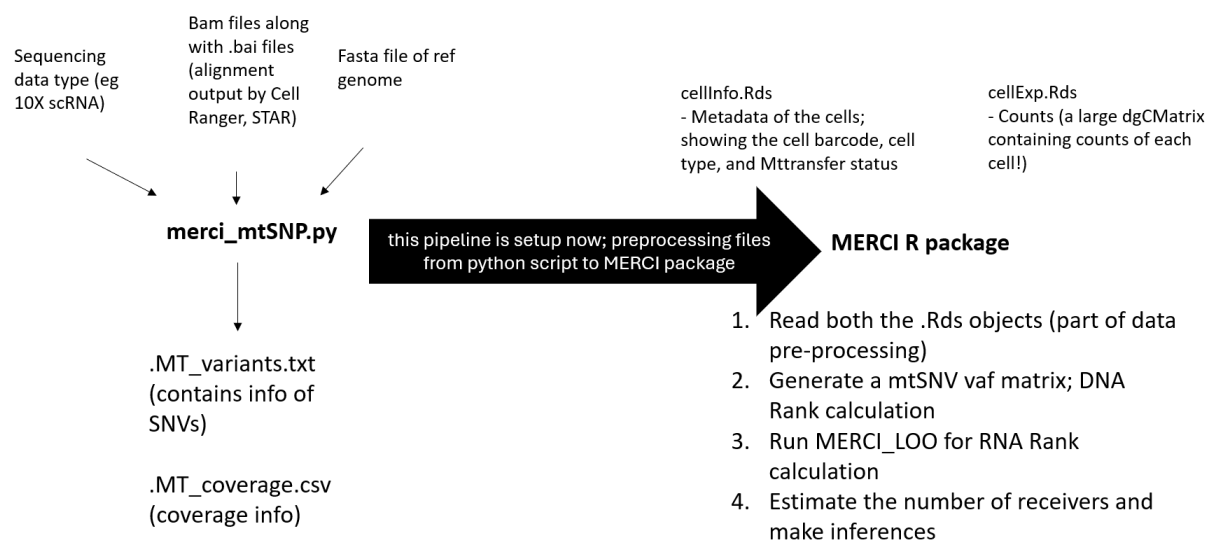


Figure 12. MERCI workflow overview